Cane Water Audit: Use in the Sugar House

This guide explains four daily water-use categories in a sugar factory: boiler make-up, imbibition and process, cooling and spray pond make-up, and washing and cleaning. Enter the quantity in litres per day (L/day).

1. Explanation of Each Parameter

Parameter	Meaning	Unit	What to include
Boiler make-up	Water added to the boiler system to replace losses and condensate that is not returned.	L/day	Include water needed to compensate for blowdown, steam losses, leaks and unrecovered or rejected condensate.
Imbibition and process	Water used in sugar extraction and other sugar-manufacturing operations included in this category.	L/day	Include imbibition water applied to bagasse and other process-water uses defined by the audit.
Cooling and spray pond make-up	Water added to cooling systems to replace evaporation, blowdown, drift, leakage and other losses.	L/day	Include make-up supplied to cooling towers, spray ponds and associated cooling circuits.
Washing and cleaning	Water used to clean machinery, equipment, floors and process areas.	L/day	Include floor washing, equipment cleaning, filter/screen washing and other plant-cleaning uses.

2. Detailed Explanation and Examples

A. Boiler Make-up (L/day)

Boilers generate steam for turbines, process heating, juice evaporation and other operations. Boiler make-up is the additional water required to maintain the system after accounting for condensate returned and other relevant flows.

Item	Illustrative quantity
Boiler-feed water requirement	10,00,000 L/day
Condensate returned	8,00,000 L/day
Additional make-up required	2,00,000 L/day

Illustrative input: Boiler make-up = 2,00,000 L/day. Actual values should be based on measured feedwater, condensate-return, blowdown and other relevant flows.

Water-saving opportunity: Recover clean condensate from evaporators and steam-heated equipment. Check its quality and divert contaminated condensate before returning water to the boiler.

B. Imbibition and Process Water (L/day)

Imbibition water is applied to bagasse during milling to improve sucrose extraction. Other process-water uses may also be included, depending on the factory configuration and the audit definitions.

Use	Illustrative quantity
Imbibition water	5,00,000 L/day
Other process water	1,00,000 L/day
Total	6,00,000 L/day

Illustrative input: Imbibition and process = 6,00,000 L/day. Use flow-meter readings or operating records. Optimise application and reuse suitable condensate, but do not reduce imbibition indiscriminately because it can affect sugar extraction efficiency.

C. Cooling and Spray Pond Make-up (L/day)

Cooling systems reject heat and lose water through evaporation, blowdown, drift and leakage. Make-up water replaces these losses.

Cooling-system loss	Illustrative quantity
Evaporation	1,50,000 L/day
Blowdown	30,000 L/day
Drift and leakage	5,000 L/day
Total make-up requirement	1,85,000 L/day

Illustrative input: Cooling and spray pond make-up = 1,85,000 L/day, assuming no other significant inflows or storage changes. Water-saving options include optimising cooling-tower cycles of concentration (COC), repairing leaks, monitoring blowdown and reusing suitable condensate or treated water. Evaporation is necessary for heat rejection; focus on avoidable losses and efficient operation.

D. Washing and Cleaning (L/day)

This category covers water used for cleaning machinery, floors, equipment and production areas.

Cleaning activity	Illustrative quantity
Floor washing	20,000 L/day
Equipment cleaning	15,000 L/day
Filter and screen washing	10,000 L/day
Other cleaning	5,000 L/day
Total	50,000 L/day

Illustrative input: Washing and cleaning = 50,000 L/day. Reduce avoidable use through dry cleaning before washing, controlled-flow hoses and trigger-operated nozzles. Reused water must meet the hygiene and process requirements of the intended application.

3. Illustrative Data for the Audit Tool

The values below are examples only, not standard consumption figures for all sugar factories.

Parameter	Illustrative input
Boiler make-up	2,00,000 L/day
Imbibition and process	6,00,000 L/day
Cooling and spray pond make-up	1,85,000 L/day
Washing and cleaning	50,000 L/day
Total for these four categories	10,35,000 L/day = 1,035 m³/day

4. How to Use These Inputs

- Record daily water use for each category using meters wherever possible.
- Separate fresh-water input from recovered or reused water when the software permits it.
- Define each category clearly to avoid counting the same water flow twice.
- Use actual factory data before calculating water-saving potential or financial benefits.
- Document whether each value is measured, calculated or estimated.